

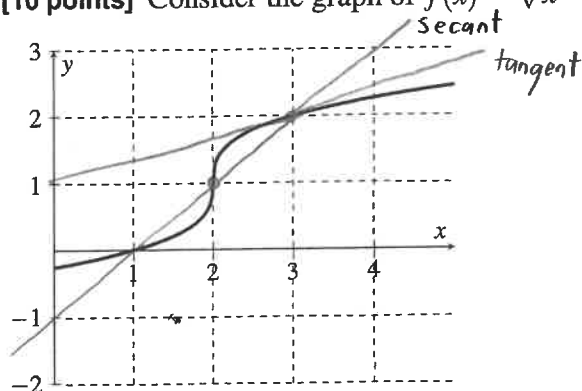
Name: Key

/ 25

Please circle your section: 901 (Gossell) 902 (Gossell) 903 (Graves-McCleary)

No aids (book, calculator, etc.) are permitted. Show all work for full credit.

1. [10 points] Consider the graph of
- $f(x) = \sqrt[3]{x-2} + 1$
- given below:



- a. Find the equation of the
- secant line**
- passing through the points (2, 1) and (3, 2).

$$m = \frac{2-1}{3-2} = 1 \quad y - 1 = 1(x - 2) \rightarrow y = x - 2 + 1 \rightarrow \boxed{y = x - 1}$$

- b. Sketch the secant line from part (a) on the graph above. Label it "Secant Line".

✓

- c. The table below lists the slope (
- m_{sec}
-) of the secant line passing through the point (3, 2) and the point
- $(x, f(x))$
- for several values of
- x
- .

x	2.9	2.99	2.999	3.001	3.01	3.1
$f(x)$	1.96549	1.99666	1.99967	2.00033	2.00333	2.03228
m_{sec}	0.34511	0.33445	0.33344	0.33322	0.33278	0.32280

Use the information in the table to **estimate** the slope of the **tangent line** to $f(x)$ at (3, 2).

$$m = \frac{1}{3}$$

- d. Use the slope from part (c) to write the equation of the tangent line at (3, 2).

$$\boxed{y - 2 = \frac{1}{3}(x - 3)} \rightarrow y = \frac{1}{3}x - 1 + 2 \rightarrow \boxed{y = \frac{1}{3}x + 1}$$

- e. Sketch the tangent line from part (d) on the graph above. Label it "Tangent Line".

✓

2. [7 points] The height of a hot air balloon after t minutes is given by $h(t) = 1500 - \frac{3000}{1+2^t}$ feet.

- a. Find the height of the hot air balloon at time $t = 0$, $t = 1$, and $t = 2$.

$$h(0) = 1500 - \frac{3000}{2} = 0$$

$$h(1) = 1500 - \frac{3000}{3} = 500$$

$$h(0) = \underline{0} \quad h(1) = \underline{500} \quad h(2) = \underline{900} \quad h(2) = 1500 - \frac{3000}{5} = 900$$

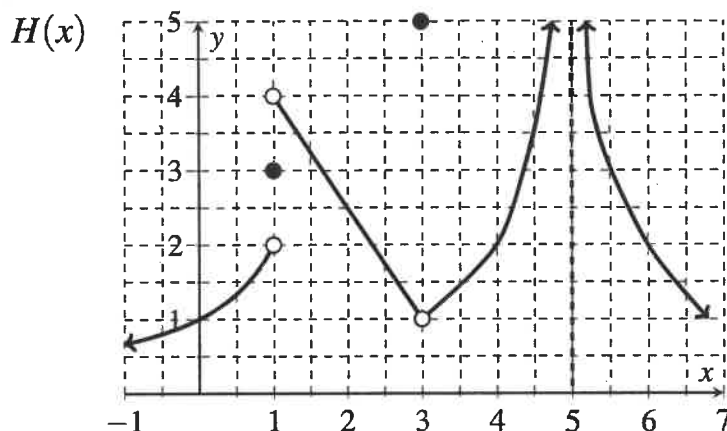
- b. Find the **average velocity** of the hot air balloon during the first minute (from $t = 0$ to $t = 1$).

$$\frac{500 - 0}{1 - 0} = 500 \text{ ft/min}$$

- c. Find the **average velocity** of the hot air balloon during the next minute (from $t = 1$ to $t = 2$).
Based on this, do you think the balloon is speeding up or slowing down?

$$\frac{900 - 500}{2 - 1} = 400 \text{ ft/min} \quad \text{The balloon is slowing down.}$$

3. [8 points] Use the graph of the function $H(x)$ (drawn below) to answer the questions. Assume $H(x)$ has a vertical asymptote at $x = 5$. For each problem below, give the most complete answer; if the limit is infinite, indicate that with $+\infty$ or $-\infty$. If the limit does not exist, write DNE.



a. $H(1) = \underline{3}$ b. $\lim_{x \rightarrow 1^-} H(x) = \underline{2}$ c. $\lim_{x \rightarrow 1^+} H(x) = \underline{4}$ d. $\lim_{x \rightarrow 1} H(x) = \underline{\text{DNE}}$

e. $\lim_{x \rightarrow 2} H(x) = \underline{2.5}$ f. $H(3) = \underline{5}$ g. $\lim_{x \rightarrow 3} H(x) = \underline{1}$ h. $\lim_{x \rightarrow 5} H(x) = \underline{+\infty}$